HISTORICAL LANGUAGE TECHNOLOGY / ULTRA-LOW-RESOURCE POST-TRAINING

Tangut-SPAR

Structured Proxy Augmentation and Reliable Preference Tuning

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400 real training pairs · Qwen2.5-7B-Instruct · Dictionary-anchored learning

01 Learning the supervision before optimizing the preference

Tangut title recovery combines an independent script, scarce parallel data and compact historical Chinese outputs. Fluent paraphrases can erase essential title structure.

Tangut-SPAR asks how synthetic evidence becomes useful supervision, and when preference tuning improves the learned mapping without destabilizing historical title recovery.

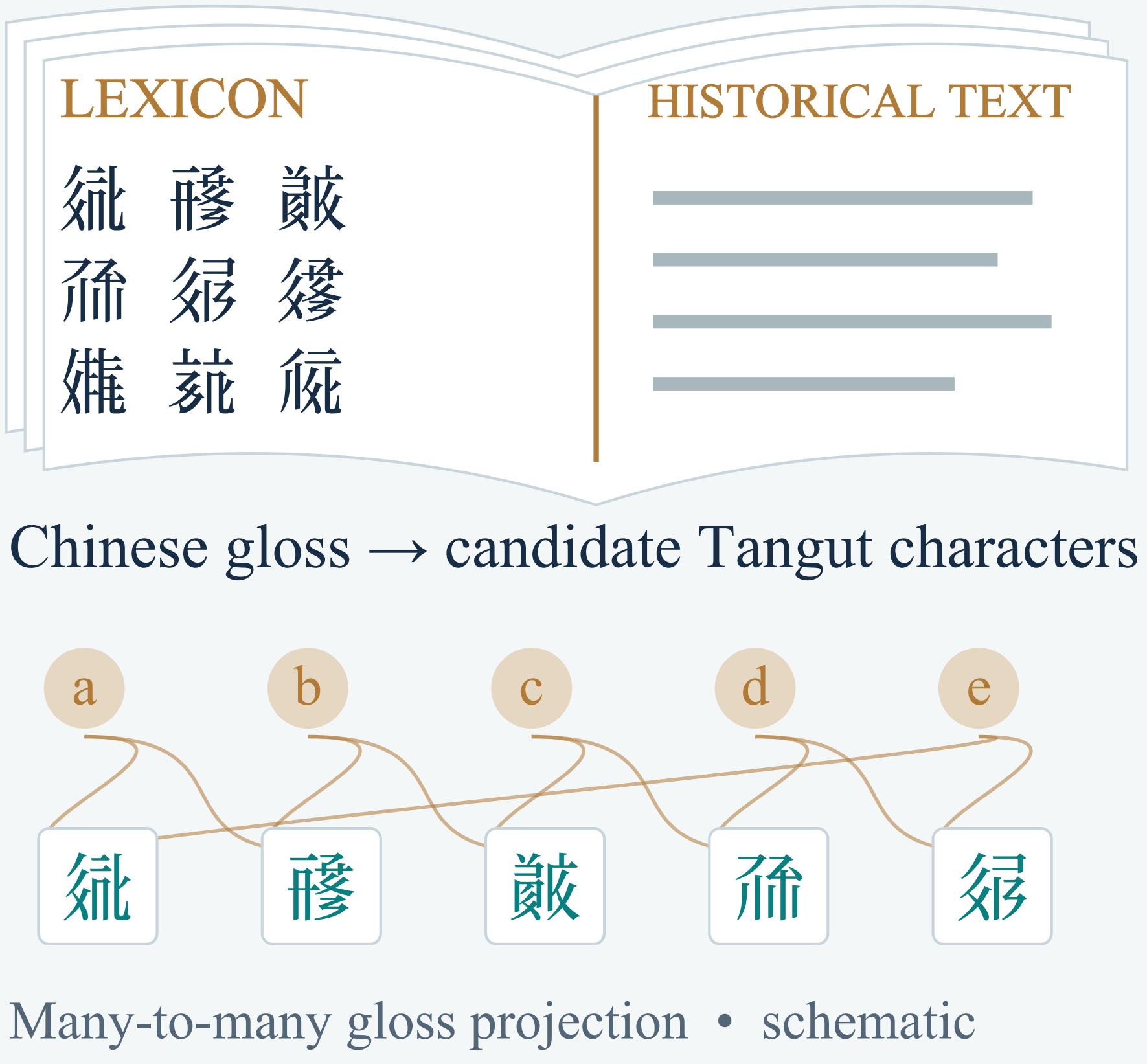
Data audit: 0 exact split overlaps; 0 synthetic-target / test-reference exact overlaps.

491
real pairs
400 train / 41 dev / 50 test

50k
proxy pairs / branch
+ 10× real-train sampling

02 A single evidence-to-preference learning system

A BUILD SYMBOLIC EVIDENCE



Chinese gloss → candidate Tangut characters

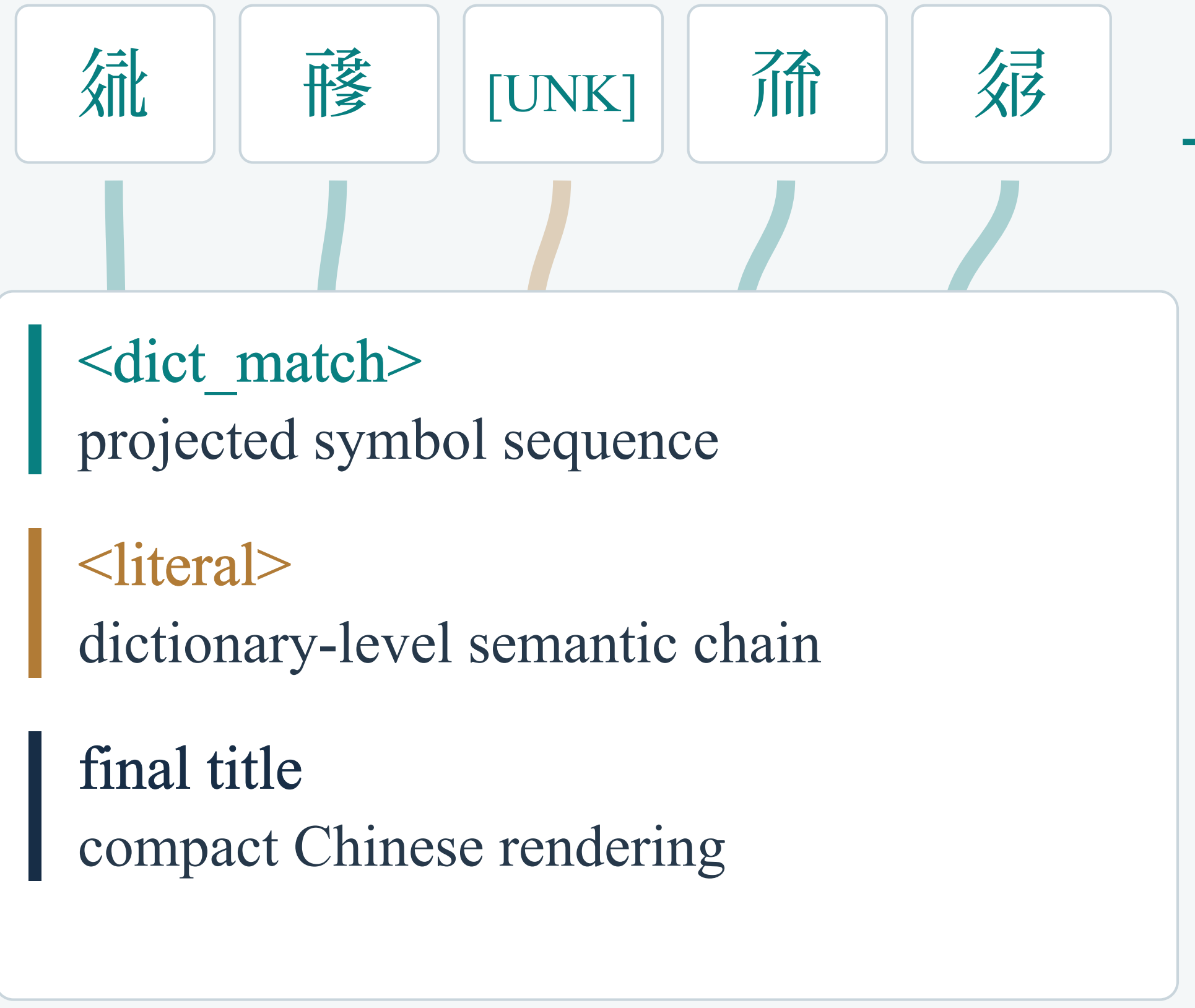
Many-to-many gloss projection • schematic

Mixed	30–70% replacement; Hanzi remain
UNK	Dictionary misses → [UNK]
Semantic	Embedding / FAISS top-3 / vote
Multitask	UNK source + structured target

400 real pairs × 10 + ≤50k proxy pairs

B EXPOSE THE ALIGNMENT

Proxy source



<dict_match>
projected symbol sequence

<literal>
dictionary-level semantic chain

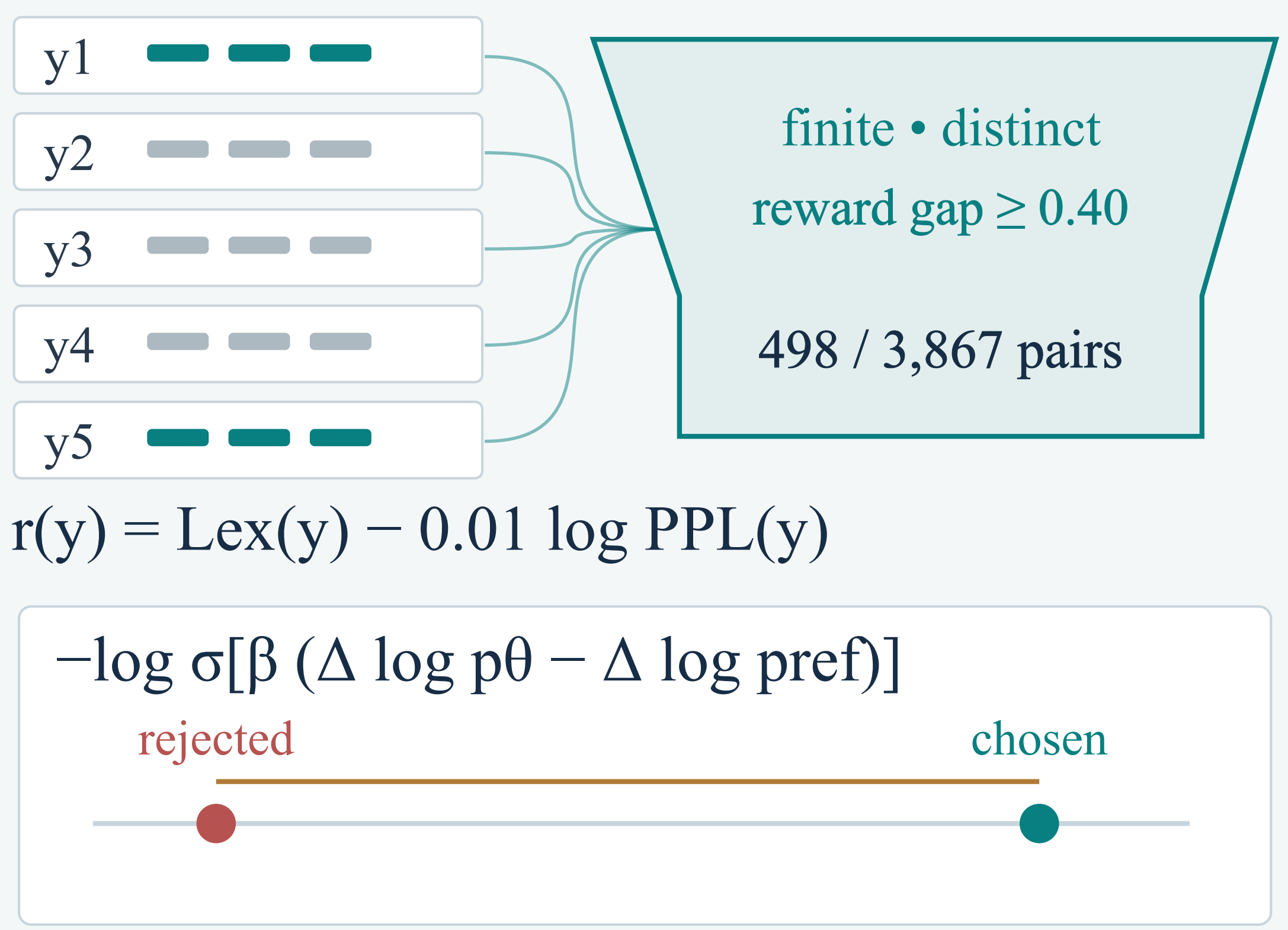
final title
compact Chinese rendering

$$L_{SFT} = -\sum \log p_\theta(z_t \mid x, z_{<t})$$

z serializes symbol evidence, a literal draft and the final target. Intermediate fields provide explicit training supervision.

C CONTROL THE UPDATE

Sample 5 candidates / prompt



$$r(y) = \mathrm{Lex}(y) - 0.01 \log \mathrm{PPL}(y)$$

$$-\log \sigma[\beta\,(\Delta \log p_\theta - \Delta \log \mathrm{pref})]$$

Δ = chosen − rejected log-probability

policy + reference ← MT-SFT

Sigmoid DPO; β = 0.1. Filter pair validity before optimizing the ranking.

Qwen2.5-7B-Instruct + LoRA
Structured SFT: rank 64, 3 epochs
→
Reliable preference tuning
DPO: rank 32, 2 epochs

HISTORICAL FAILURE
Different initialization

Earlier SFT base + weak / invalid preference pairs → legacy DPO
19.39 chrF++ · 18% contamination · 1.88× reference length

03 Recovery is structural, not just exact memorization

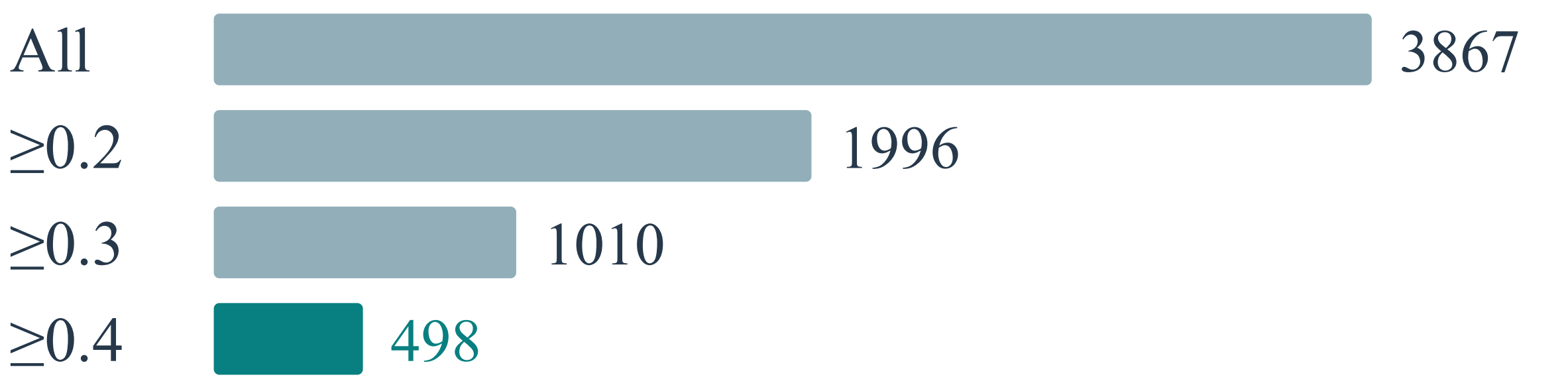
Local learned model	chrF++ ↑	Exact ↑	Judge ↑	Suffix ↑	Noise ↓
UNK-SFT	22.11	3/50	1.82	17/27	6%
MT-SFT	25.99	5/50	1.98	19/27	2%
Legacy DPO*	19.39	2/50	1.68	18/27	18%
MT-DPO · gap 0.4	30.01	5/50	2.22	22/27	4%

+4.02 chrF++ over MT-SFT; exact match stays at 5/50. More title structure is recovered among non-exact outputs.

50 held-out titles. Judge: reference-aware overall / 5. Suffix: 27 eligible titles. Noise: Latin / markup / prompt artifacts. *Historical control uses a different base.

04 Confidence trades volume for signal

Pairs retained as the reward gap increases



71.3% → 81.5% 12.9% kept

Chosen closer to a target proxy: all → gap ≥ 0.4. Proxy agreement measures preference-label alignment.

Offline reward audit	Keep ≥0.2	Good @0.4
Lex only	1,994	81.8%
Literal only	166	88.0%
½ lex + ½ literal	832	86.8%

05 Read the recovery in real Tangut titles

TERMINOLOGICAL CHAIN

[illegible]

Reference	正理滴之句義顯具
MT-SFT	正理滴之要義顯註
MT-DPO	正理滴之句義顯了

Restores “句義”; the final lexical choice remains imperfect.

LONG-TITLE RECOVERY

[illegible]

Reference	勝慧彼岸到要論教學現量解莊嚴論顯頌
MT-SFT	勝慧到彼岸要論　教學現量解莊嚴顯疏
MT-DPO	勝慧彼岸到要論教學現量解莊嚴顯頌

Preserves the long chain; one reference “論” is still missing.

06 What the diagnosis tells us

Supervision structure [1]

Intermediate targets expose evidence before generation. MT-SFT gains 3.88 chrF++ over UNK-SFT and cuts contamination from 6% to 2%, consistent with a useful alignment scaffold.

Preference validity [2, 3]

A generic LM assigns human references PPL ≈ 1.65M. Fluency and historical title fidelity diverge; noisy preference labels can reinforce the wrong generation behavior.

Loss complexity [4]

Robust + WPO did not beat sigmoid on chrF++ at gap 0.4. Reweighting off-policy data does not itself establish correct preference labels.

[1] Hsieh et al., 2023 · arXiv:2305.02301 [2] Rafailov et al., 2023 · arXiv:2305.18290
[3] Fisch et al., 2024 · arXiv:2405.19316 [4] Zhou et al., 2024 · arXiv:2406.11827

Project evidence: github.com/ChouYuanjue/tangut-nlp
Short historical titles · reported experiments · no new training runs